THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 66°	Sum is 90°.
2	**(d)** 60°	Total parts $= k + 1$.
3	**(a)** 115°	Co-interior angles are supplementary.
4	**(c)** parallel	Use the property of parallel lines.
5	**(b)** 28°	Vertically opposite angles are equal.
6	**(c)** acute	Classify by measure.
7	**(a)** 90°	Total is 360°.
8	**(b)** 53°	Sum is 90°.
9	**(d)** 36°	Total parts $= k + 1$.
10	**(a)** 100°	Co-interior angles are supplementary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2x + 10 + 70 = 180$ (1); $x = 50$ (1).	1+1
12	Alternate interior angles are equal (1); $x + 10 = 65$, so $x = 55$ (1).	1+1
13	$x + 2x = 180$ (1); $x = 60^\circ$, other 120° (1).	1+1
14	$2x + 20 + 100 = 180$ (1); $x = 30$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Corresponding angles equal (1); $3x + 5 = 2x + 40$ so $x = 35$ (1); Angle $= 110^\circ$ (1).	1+1+1
16	Sum is 180° (1); $6x = 162$ so $x = 27$ (1); Angles 114° and 66° (1).	1+1+1
17	Vertically opposite angles are equal; $5x - 10 = 3x + 20$ so $x = 15$ (1); Equal angles 65° each (1); Other two 115° each (1).	1+1+1
	OR: Corresponding angles equal (1); $3x + 10 = 2x + 55$ so $x = 45$ (1); Angle $= 145^\circ$ (1).	1+1+1
18	Sum is 180° (1); $6x = 150$ so $x = 25$ (1); Angles 110° and 70° (1).	1+1+1
	OR: Vertically opposite angles are equal; $5x - 6 = 3x + 14$ so $x = 10$ (1); Equal angles 44° each (1); Other two 136° each (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Corresponding $= 50^\circ$ (1); Alternate interior $= 50^\circ$ (it is vertically opposite to the corresponding angle) (1); Vertically opposite to $\angle 1 = 50^\circ$ (1); Co-interior $= 130^\circ$ (1); Give a reason for each (1).	1+1+1+1+1
	OR: Equal to 50°: vertically opposite, corresponding and alternate angles (4 angles) (1); Remaining 4 angles form linear pairs with it, so 130° each (1); Linear pair: sum 180° (1); Co-interior pairs are supplementary (1); Check: $4(50) + 4(130) = 720^\circ$ (1).	1+1+1+1+1
20	Corresponding $= 65^\circ$ (1); Alternate interior $= 65^\circ$ (it is vertically opposite to the corresponding angle) (1); Vertically opposite to $\angle 1 = 65^\circ$ (1); Co-interior $= 115^\circ$ (1); Give a reason for each (1).	1+1+1+1+1
	OR: Equal to 65°: vertically opposite, corresponding and alternate angles (4 angles) (1); Remaining 4 angles form linear pairs with it, so 115° each (1); Linear pair: sum 180° (1); Co-interior pairs are supplementary (1); Check: $4(65) + 4(115) = 720^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** parallel	Use the property of parallel lines.
2	**(d)** 64°	Vertically opposite angles are equal.
3	**(a)** right	Classify by measure.
4	**(d)** 110°	Total is 360°.
5	**(c)** 39°	Sum is 90°.
6	**(c)** 30°	Total parts $= k + 1$.
7	**(b)** 68°	Co-interior angles are supplementary.
8	**(a)** not parallel	Use the property of parallel lines.
9	**(d)** 97°	Vertically opposite angles are equal.
10	**(c)** obtuse	Classify by measure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Alternate interior angles are equal (1); $x + 25 = 80$, so $x = 55$ (1).	1+1
12	$x + 3x = 180$ (1); $x = 45^\circ$, other 135° (1).	1+1
13	$2x + 30 + 80 = 180$ (1); $x = 35$ (1).	1+1
14	Alternate interior angles are equal (1); $x + 15 = 110$, so $x = 95$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Corresponding angles equal (1); $3x + 15 = 2x + 60$ so $x = 45$ (1); Angle $= 150^\circ$ (1).	1+1+1
16	Sum is 180° (1); $6x = 150$ so $x = 25$ (1); Angles 112° and 68° (1).	1+1+1
17	Vertically opposite angles are equal; $5x - 8 = 3x + 18$ so $x = 13$ (1); Equal angles 57° each (1); Other two 123° each (1).	1+1+1
	OR: Corresponding angles equal (1); $3x + 8 = 2x + 45$ so $x = 37$ (1); Angle $= 119^\circ$ (1).	1+1+1
18	Sum is 180° (1); $6x = 162$ so $x = 27$ (1); Angles 112° and 68° (1).	1+1+1
	OR: Vertically opposite angles are equal; $5x - 12 = 3x + 30$ so $x = 21$ (1); Equal angles 93° each (1); Other two 87° each (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Corresponding $= 75^\circ$ (1); Alternate interior $= 75^\circ$ (it is vertically opposite to the corresponding angle) (1); Vertically opposite to $\angle 1 = 75^\circ$ (1); Co-interior $= 105^\circ$ (1); Give a reason for each (1).	1+1+1+1+1
	OR: Equal to 75°: vertically opposite, corresponding and alternate angles (4 angles) (1); Remaining 4 angles form linear pairs with it, so 105° each (1); Linear pair: sum 180° (1); Co-interior pairs are supplementary (1); Check: $4(75) + 4(105) = 720^\circ$ (1).	1+1+1+1+1
20	Corresponding $= 110^\circ$ (1); Alternate interior $= 110^\circ$ (it is vertically opposite to the corresponding angle) (1); Vertically opposite to $\angle 1 = 110^\circ$ (1); Co-interior $= 70^\circ$ (1); Give a reason for each (1).	1+1+1+1+1
	OR: Equal to 110°: vertically opposite, corresponding and alternate angles (4 angles) (1); Remaining 4 angles form linear pairs with it, so 70° each (1); Linear pair: sum 180° (1); Co-interior pairs are supplementary (1); Check: $4(110) + 4(70) = 720^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 70°	Total is 360°.
2	**(a)** 22°	Sum is 90°.
3	**(d)** 20°	Total parts $= k + 1$.
4	**(b)** 45°	Co-interior angles are supplementary.
5	**(c)** 8	Use the property of parallel lines.
6	**(a)** 143°	Vertically opposite angles are equal.
7	**(c)** reflex	Classify by measure.
8	**(c)** 90°	Total is 360°.
9	**(a)** 7°	Sum is 90°.
10	**(b)** acute	Classify by measure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x + 4x = 180$ (1); $x = 36^\circ$, other 144° (1).	1+1
12	$2x + 6 + 94 = 180$ (1); $x = 40$ (1).	1+1
13	Alternate interior angles are equal (1); $x + 30 = 95$, so $x = 65$ (1).	1+1
14	$x + 5x = 180$ (1); $x = 30^\circ$, other 150° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Corresponding angles equal (1); $3x + 20 = 2x + 70$ so $x = 50$ (1); Angle $= 170^\circ$ (1).	1+1+1
16	Sum is 180° (1); $6x = 162$ so $x = 27$ (1); Angles 116° and 64° (1).	1+1+1
17	Vertically opposite angles are equal; $5x - 4 = 3x + 16$ so $x = 10$ (1); Equal angles 46° each (1); Other two 134° each (1).	1+1+1
	OR: Corresponding angles equal (1); $3x + 12 = 2x + 50$ so $x = 38$ (1); Angle $= 126^\circ$ (1).	1+1+1
18	Sum is 180° (1); $6x = 150$ so $x = 25$ (1); Angles 106° and 74° (1).	1+1+1
	OR: Vertically opposite angles are equal; $5x - 2 = 3x + 12$ so $x = 7$ (1); Equal angles 33° each (1); Other two 147° each (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Corresponding $= 125^\circ$ (1); Alternate interior $= 125^\circ$ (it is vertically opposite to the corresponding angle) (1); Vertically opposite to $\angle 1 = 125^\circ$ (1); Co-interior $= 55^\circ$ (1); Give a reason for each (1).	1+1+1+1+1
	OR: Equal to 125°: vertically opposite, corresponding and alternate angles (4 angles) (1); Remaining 4 angles form linear pairs with it, so 55° each (1); Linear pair: sum 180° (1); Co-interior pairs are supplementary (1); Check: $4(125) + 4(55) = 720^\circ$ (1).	1+1+1+1+1
20	Corresponding $= 140^\circ$ (1); Alternate interior $= 140^\circ$ (it is vertically opposite to the corresponding angle) (1); Vertically opposite to $\angle 1 = 140^\circ$ (1); Co-interior $= 40^\circ$ (1); Give a reason for each (1).	1+1+1+1+1
	OR: Equal to 140°: vertically opposite, corresponding and alternate angles (4 angles) (1); Remaining 4 angles form linear pairs with it, so 40° each (1); Linear pair: sum 180° (1); Co-interior pairs are supplementary (1); Check: $4(140) + 4(40) = 720^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.